Factor Analysis of Work Engagement

Women & Workforce Engagement

EFA & CFA of the 9-Item Utrecht Work Engagement Scale (UWES-9)
in a Multi-Occupational Female Sample

Primary Study:

Willmer, M., Westerberg Jacobson, J. & Lindberg, M. (2019)
Frontiers in Psychology, 10, Article 2771
DOI: 10.3389/fpsyg.2019.02771

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SSIE-605: Applied Multivariate Data Analysis
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Agenda

Introduction

Work Engagement & Women in the Workforce

What is Work Engagement?

A positive, work-related state of mind characterized by three dimensions:

Vigor: High energy, mental resilience, willingness to invest effort
Dedication: Sense of significance, enthusiasm, inspiration, pride
Absorption: Being fully engrossed in work, time passes quickly

Why Study Women Specifically?

- Most UWES-9 studies use mixed-gender samples with majority male participants
- Gender may influence how work engagement is structured and experienced
- Only one prior Swedish study existed, using 186 IT consultants (63% male)
- Understanding women's engagement patterns is critical for retention & policy

(Schaufeli et al., 2002; Hallberg & Schaufeli, 2006)

Literature Review

The UWES-9 Debate: One Factor vs. Three Factors

The UWES-9 Instrument:

- 9 items rated on 0-6 scale
  (Never to All the time)
- 3 theoretical subscales:
  Vigor (3 items: 1, 2, 5)
  Dedication (3 items: 3, 4, 7)
  Absorption (3 items: 6, 8, 9)

The Core Debate:

Kulikowski (2017) review of 21 studies:
- 3 confirmed 1-factor
- 3 confirmed 3-factor
- 4 found both equivalent
- 1 supported neither

No definitive conclusion reached

Key Prior Findings:

Swedish (Hallberg & Schaufeli, 2006):
n = 186 IT consultants (37% women)
Both 1-factor & 3-factor fit equally
RMSEA = 0.13, CFI = 0.97

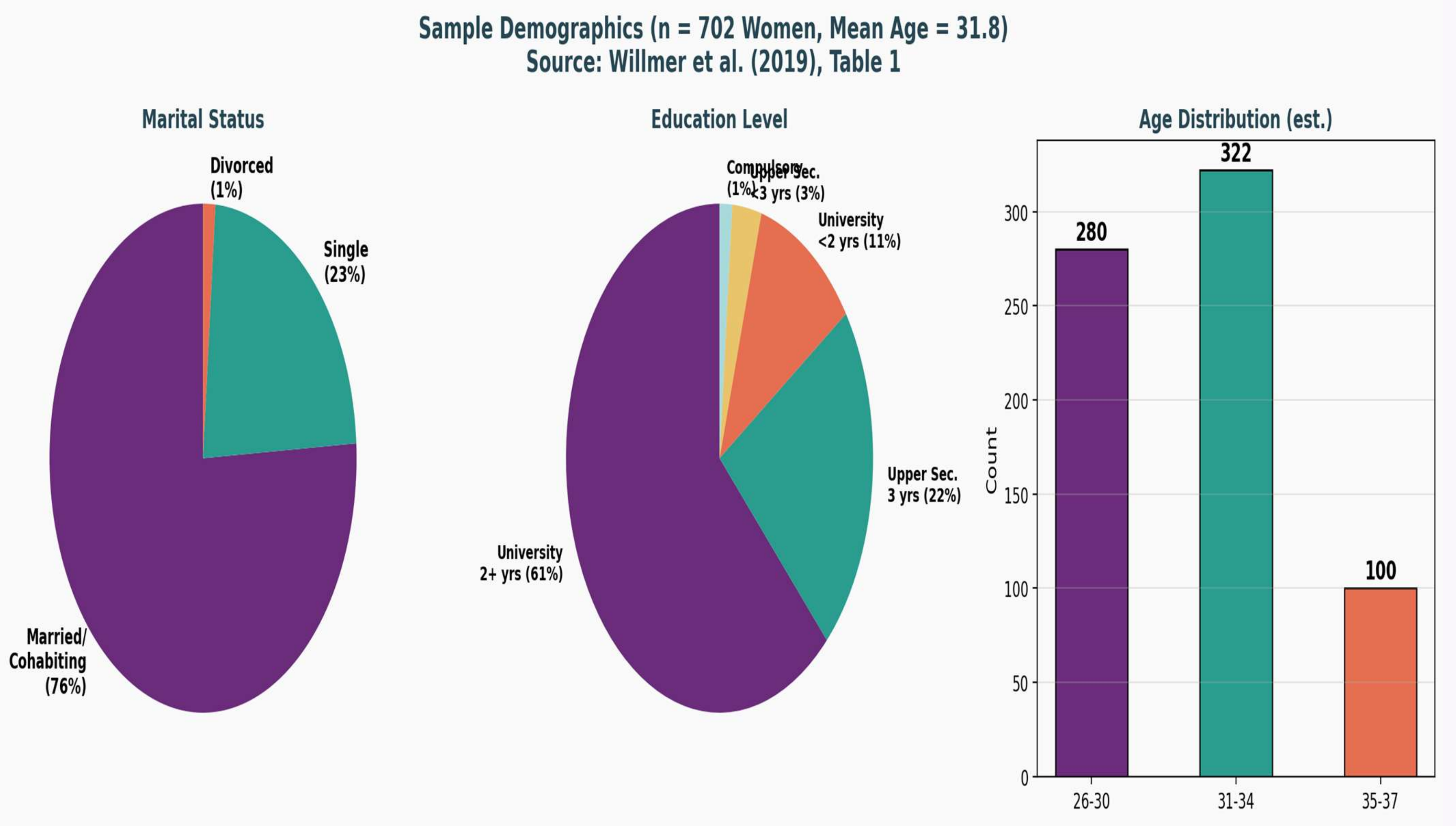
Norwegian (Nerstad et al., 2010):
n = 1,266 multi-occupational (67% women)
3-factor supported, but factors
highly correlated (suggesting 1 factor)

Finnish (Seppala et al., 2009):
n = 9,404 multi-occupational
Both structures reasonable

Gap: No all-female, multi-occupational Swedish study existed

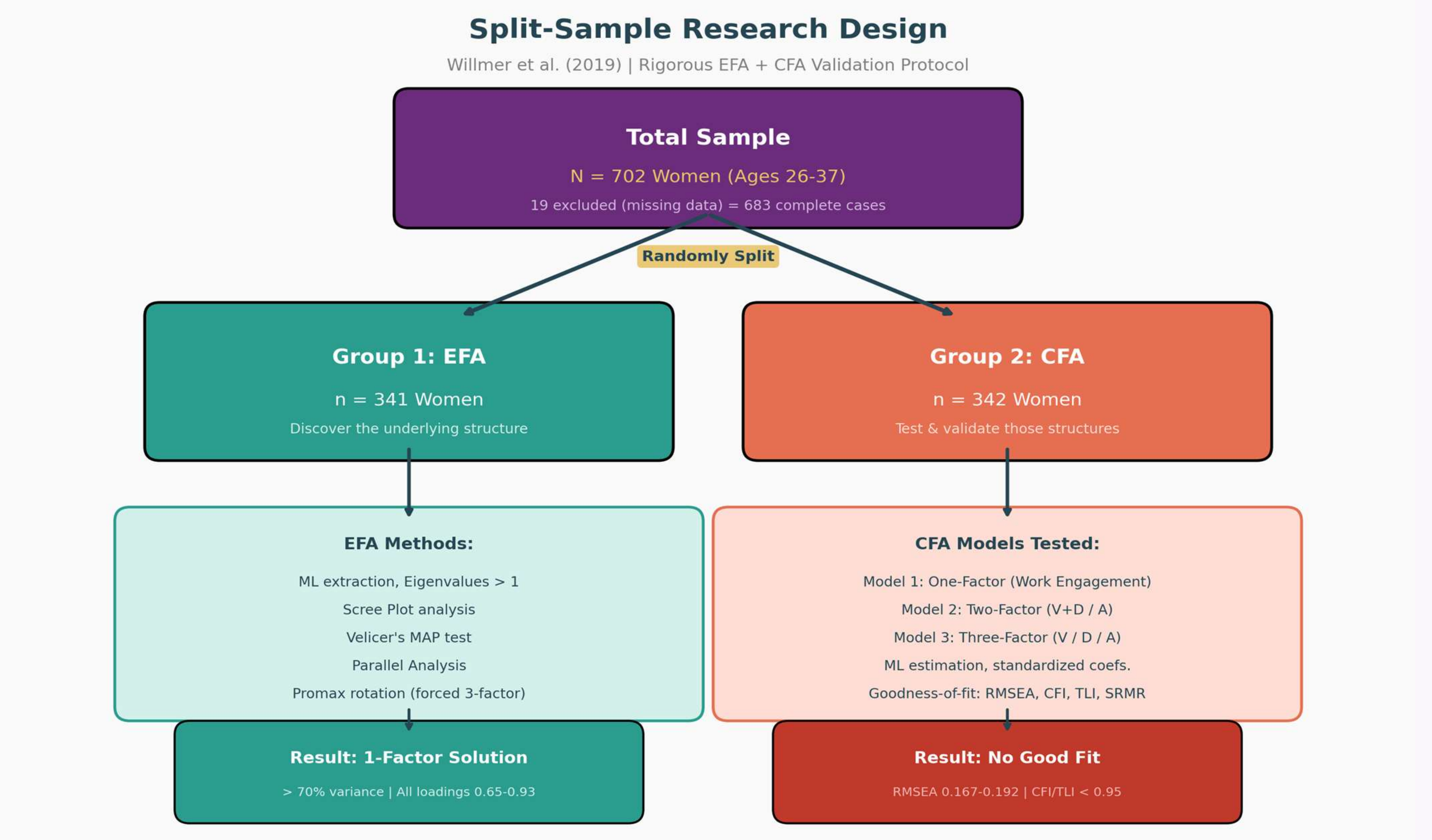
Sample Demographics

n = 702 Women, Aged 26-37, Multi-Occupational Swedish Sample



Research Methodology

Split-Sample Design: EFA + CFA



Sampling Adequacy & Reliability

KMO = 0.922 (Marvellous) | Cronbach's Alpha = 0.947

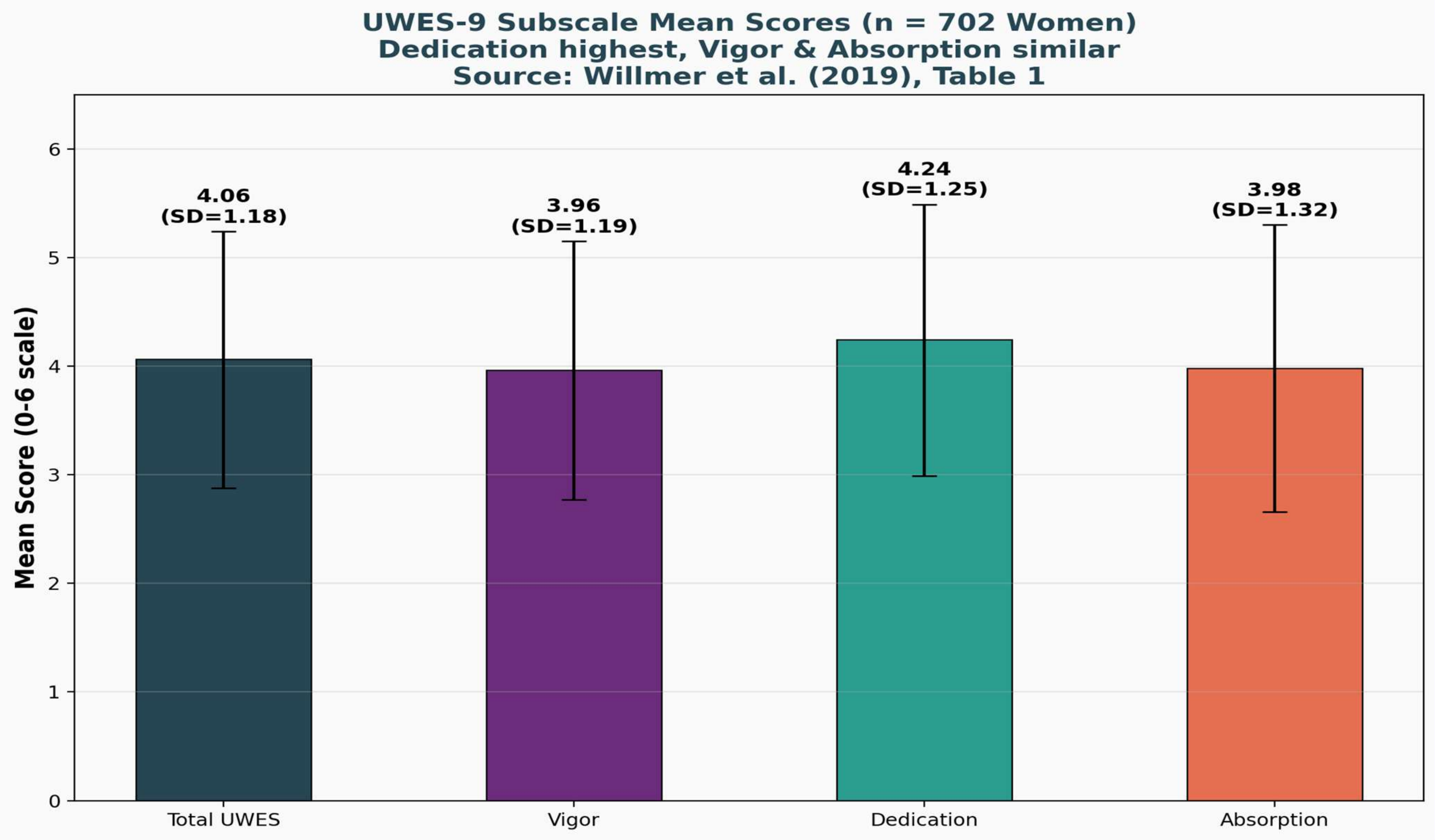
Sampling Adequacy & Sphericity Tests

Test	Value	Interpretation
KMO Measure	0.922	Marvellous (> 0.90)
Bartlett's Test p-value	< 0.001	Significant
Cronbach's Alpha	0.947	Excellent (> 0.90)
Inter-item Correlation	0.524 - 0.849	High
Subscale Correlation (V-D-A)	0.79 - 0.84	Very High
EFA Variance Explained	> 70%	One factor dominant
EFA Chi-squared (df=27)	332.43	Significant

Source: Willmer et al. (2019) | n = 702 women | UWES-9

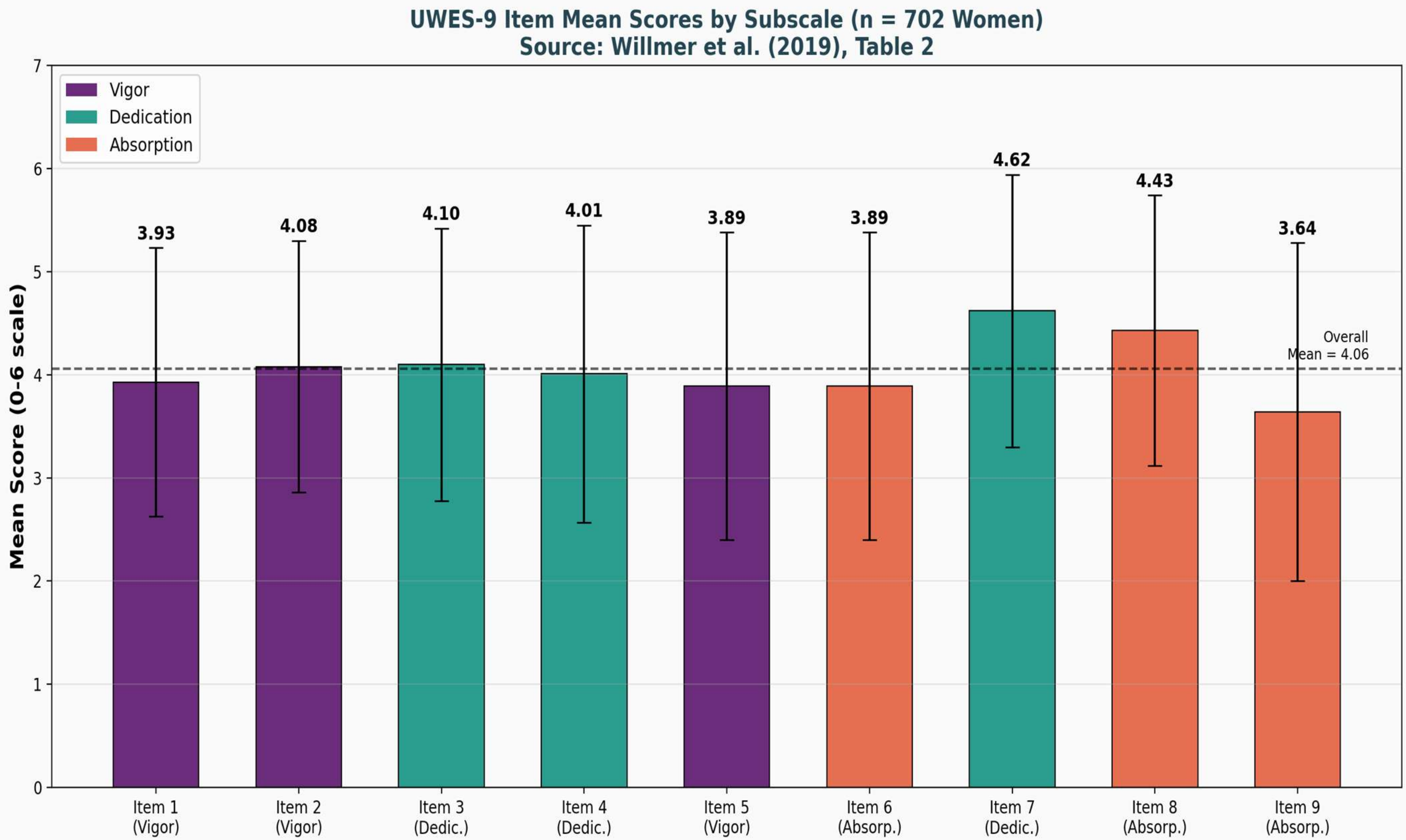
UWES-9 Subscale Scores

Overall Mean = 4.06 | Dedication Highest at 4.24



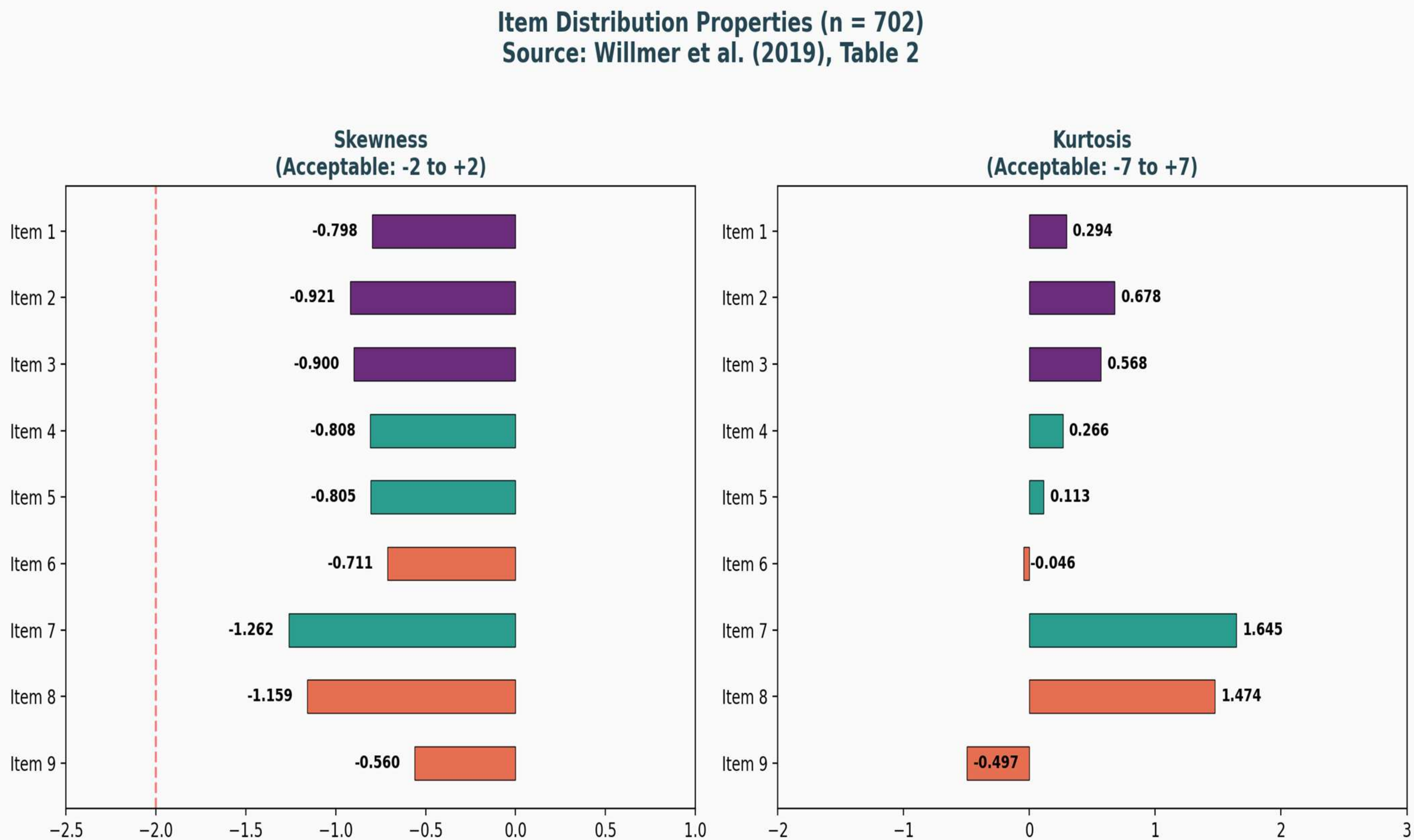
Individual Item Mean Scores

Item 7 (Pride) Highest | Item 9 (Immersion) Lowest



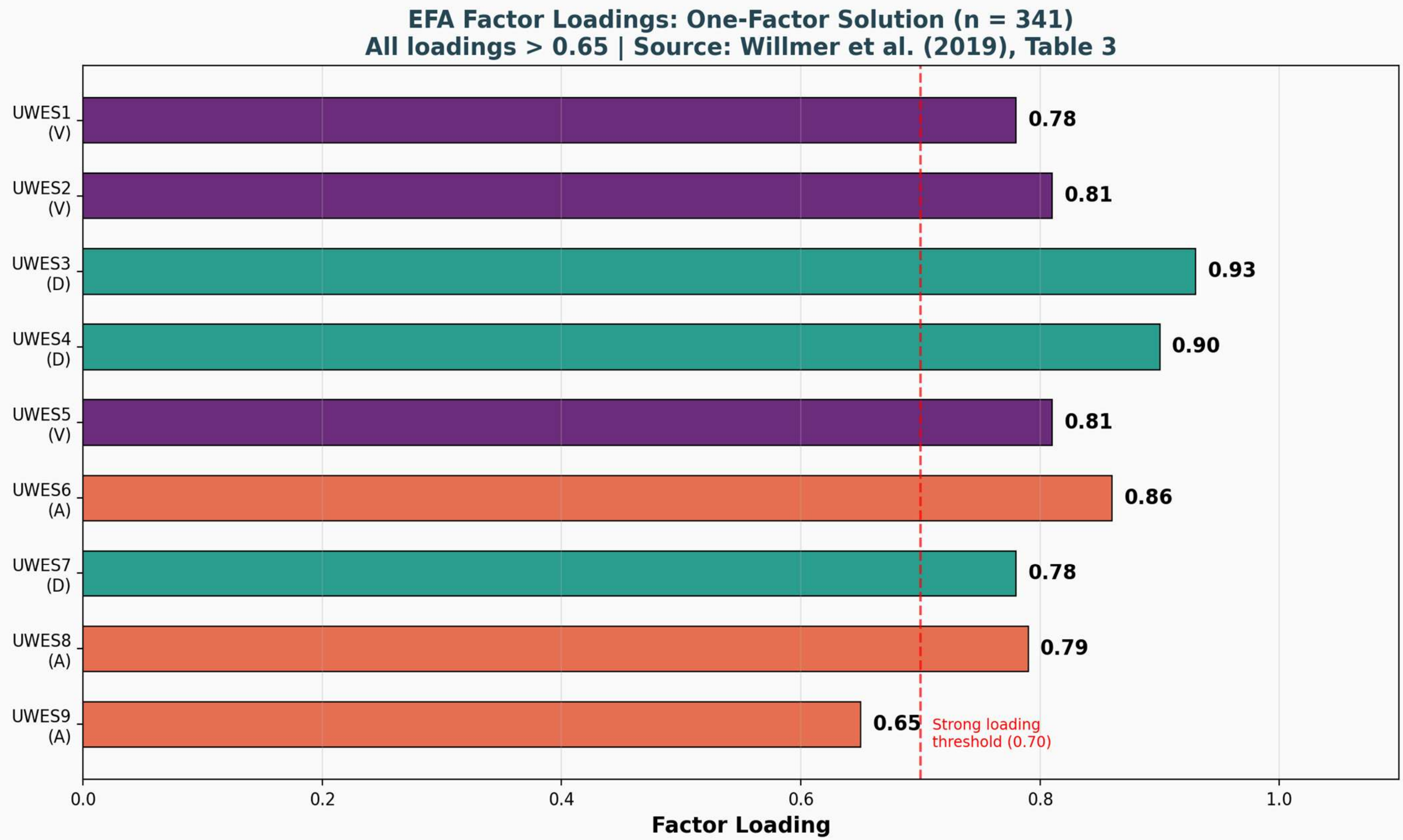
Item Distribution Properties

Skewness & Kurtosis Within Acceptable Range for ML Estimation



EFA Results: One-Factor Solution

All Loadings > 0.65 | One Factor Explains > 70% Variance



EFA Summary

Multiple Tests Converge on One-Factor Solution

EFA Key Findings (n = 341):

1. Unrotated EFA: One factor explains > 70% of variance
   - Chi-squared = 332.43 (df = 27), $p < 0.001$
2. Scree Plot: Strongly favors one-factor structure
3. Velicer's MAP Test: Both original and revised versions point to one factor
4. Parallel Analysis: Raw data eigenvalue exceeded 95th percentile for 4 factors
   - This was the only test that disagreed with the one-factor solution
5. Forced 3-Factor Promax Rotation:
   - Chi-squared = 45.72 (df = 12), $p < 0.001$
   - Items did NOT load on their expected theoretical factors
   - Dedication had 4 items (3, 4, 5, 6) instead of 3
   - Vigor had only 2 items (1, 2) instead of 3

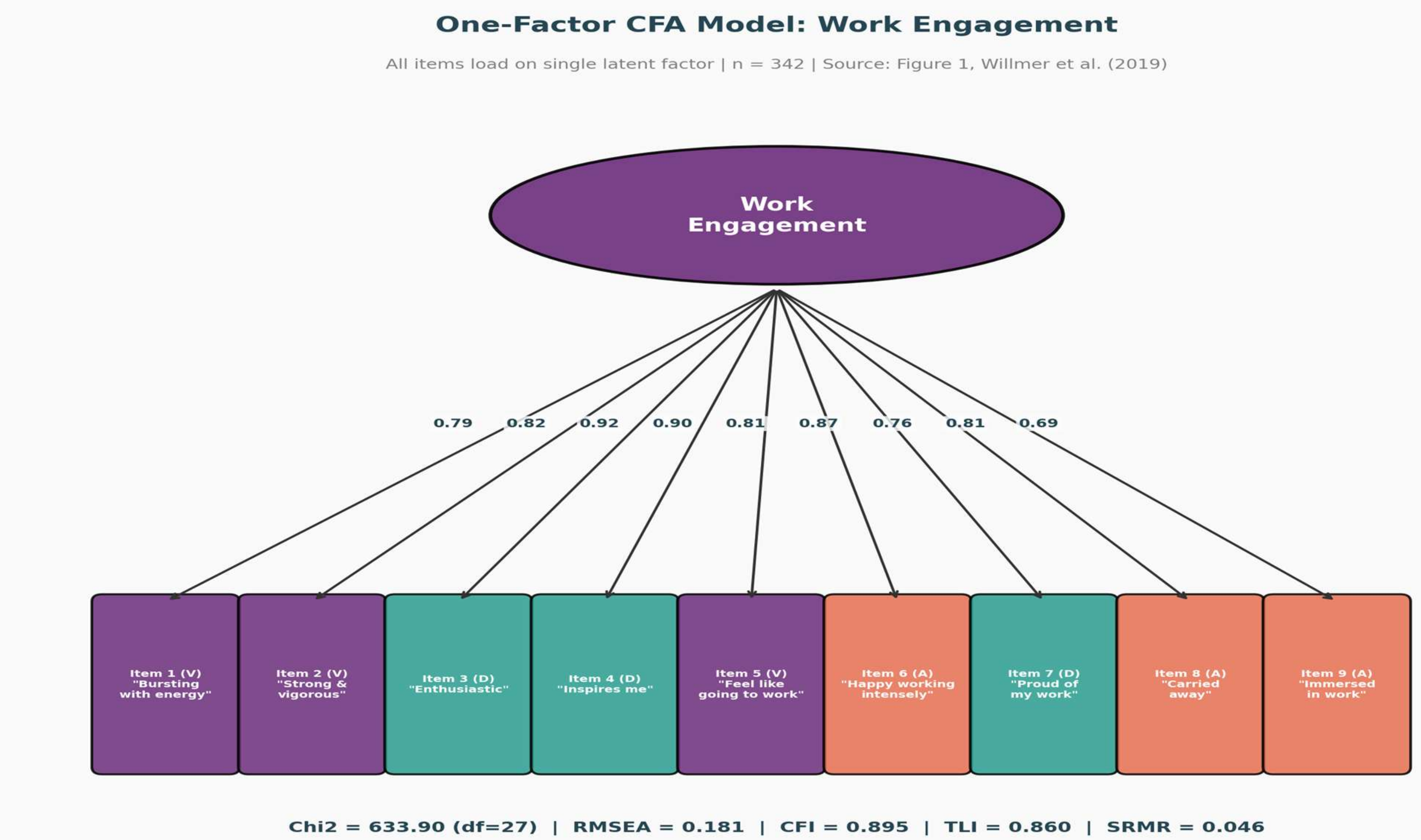
Conclusion: EFA strongly supports a single 'Work Engagement' factor

Technical Commentary:

While EFA and MAP tests supported a 1-factor solution, Parallel Analysis suggested 4 factors. This often occurs in multivariate data with high inter-item correlations (0.52-0.85), leading to 'over-factoring' -- a known limitation of Parallel Analysis in highly correlated datasets.

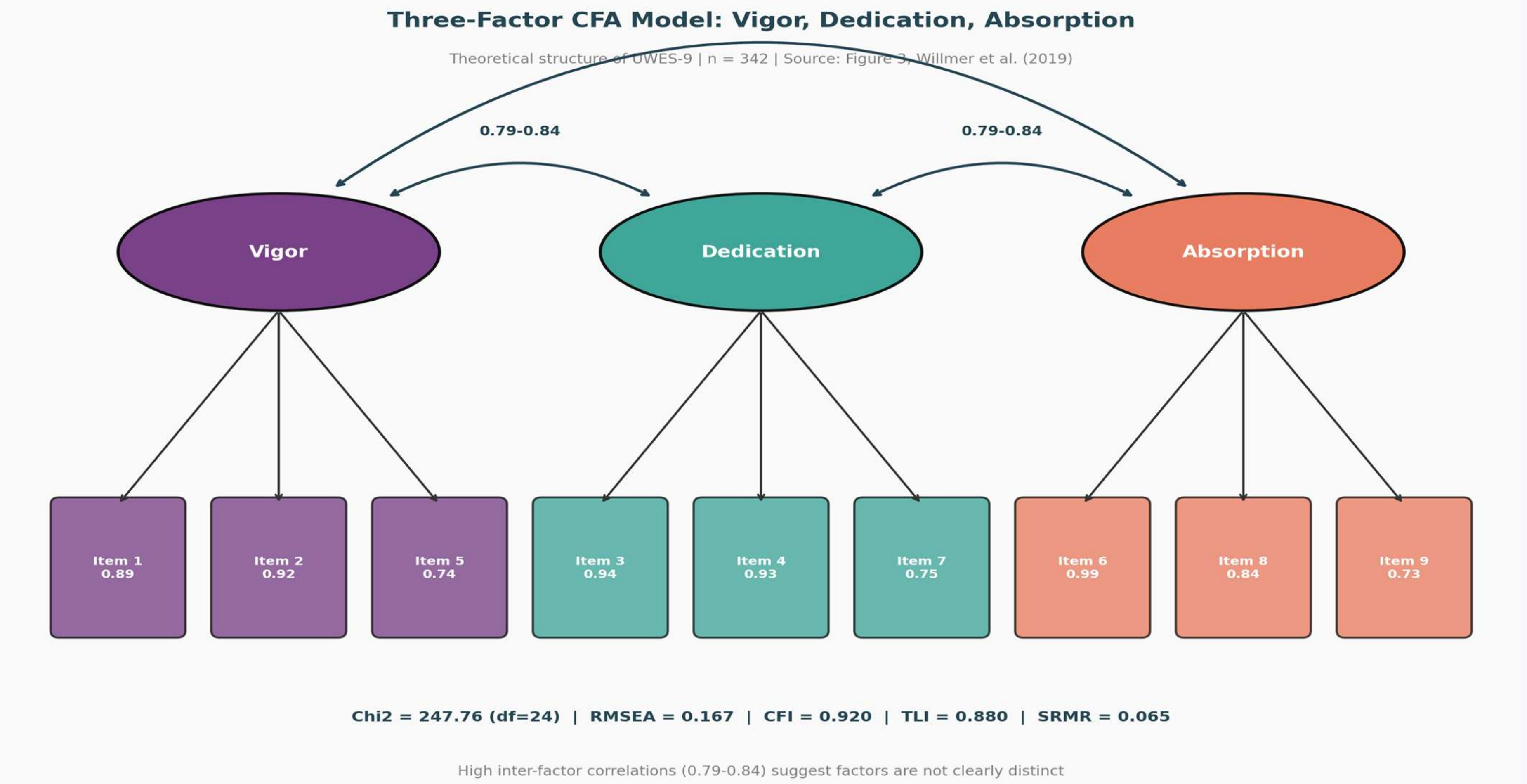
CFA Model 1: One-Factor Structure

Single Latent Factor: Work Engagement (Figure 1)



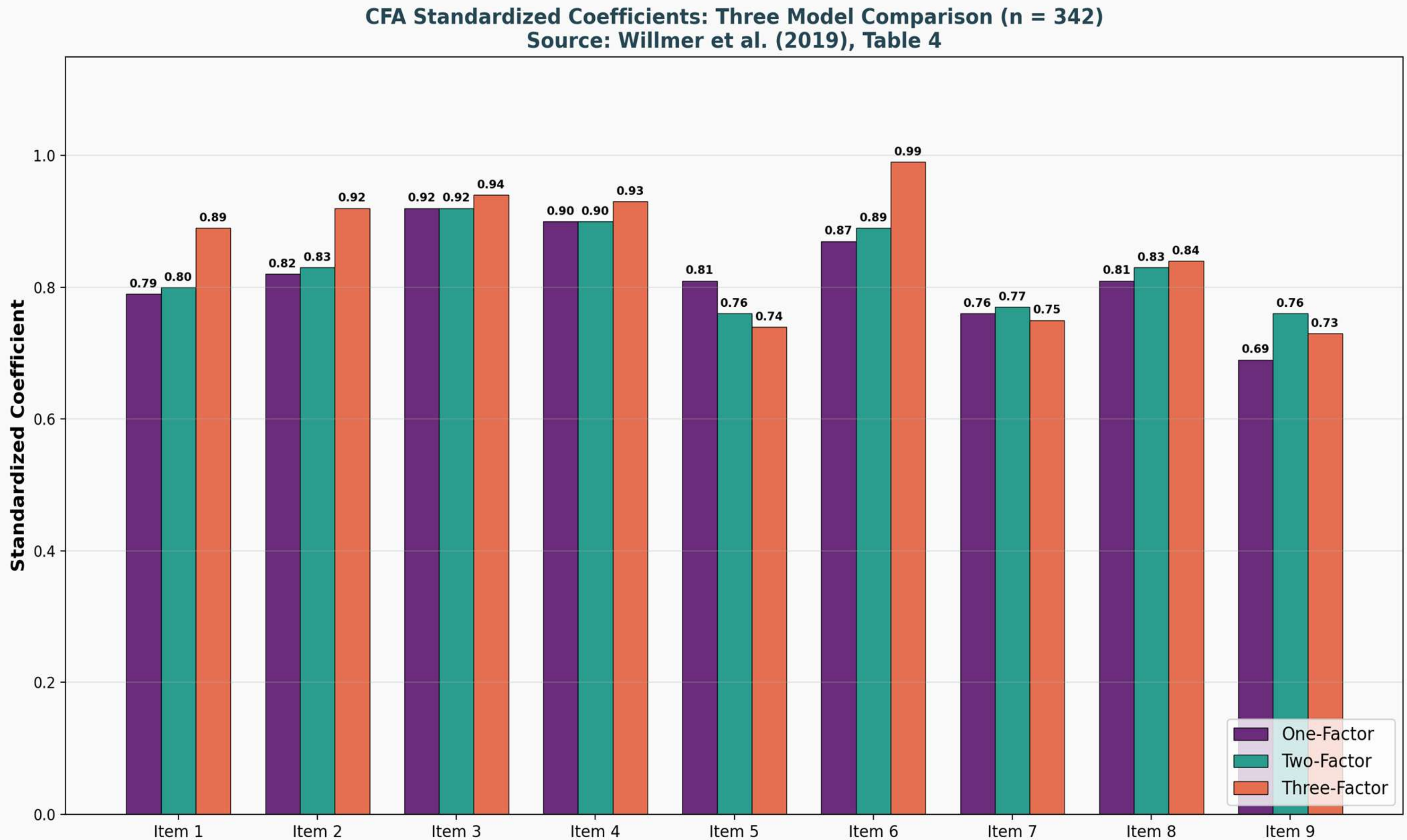
CFA Model 3: Three-Factor Structure

Vigor, Dedication, Absorption (Figure 3)



CFA Standardized Coefficients

All Three Models Compared (Table 4)



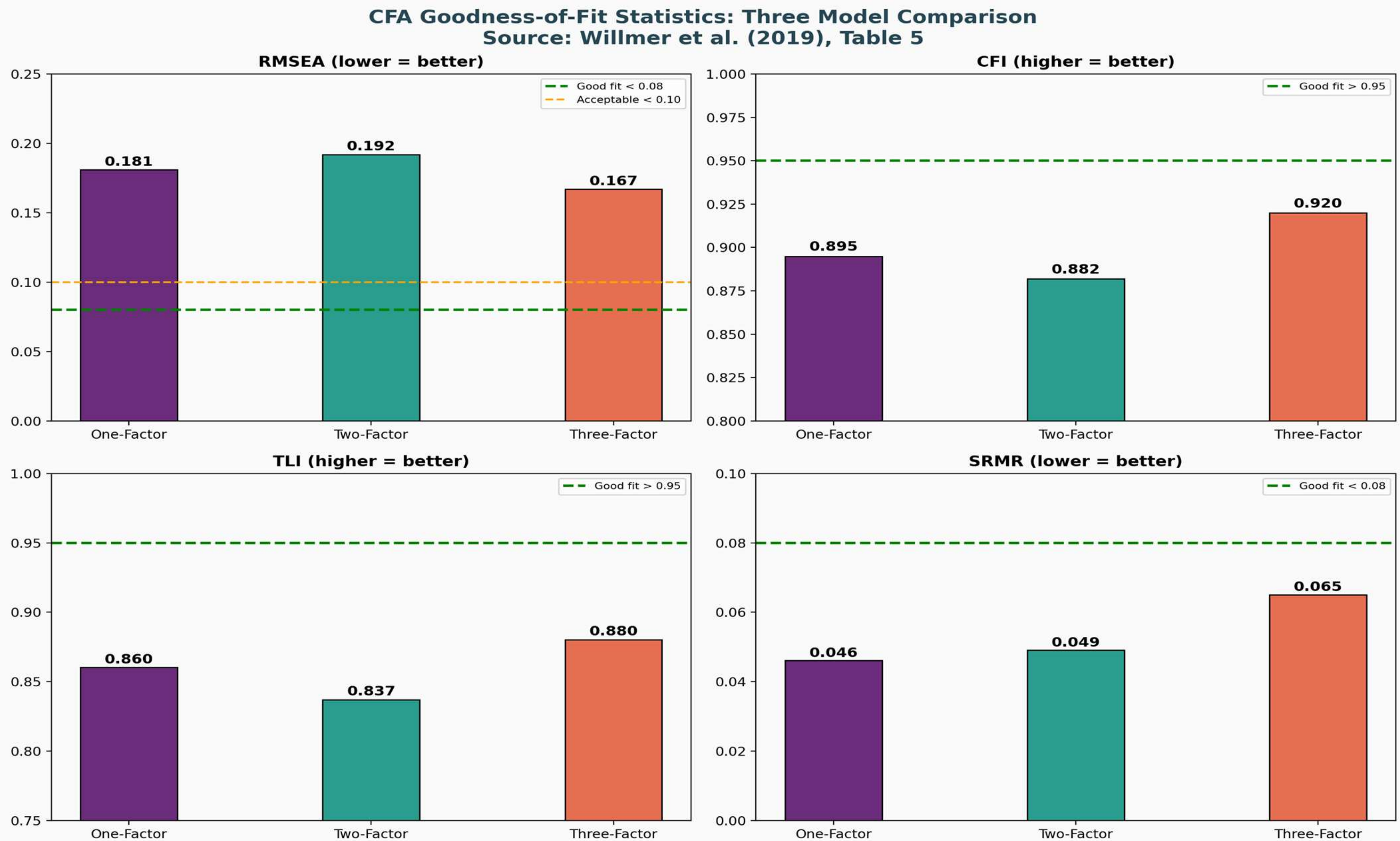
CFA Coefficients: Detailed Results

All p-values < 0.0001 (Table 4)

Item	Subscale	1-Factor Coef.	2-Factor Coef.	3-Factor Coef.	z-value (1F)
Item 1	Vigor	0.79	0.80	0.89	50.42
Item 2	Vigor	0.82	0.83	0.92	59.95
Item 3	Dedication	0.92	0.92	0.94	132.99
Item 4	Dedication	0.90	0.90	0.93	109.15
Item 5	Vigor	0.81	0.76	0.74	55.85
Item 6	Absorption	0.87	0.89	0.99	83.55
Item 7	Dedication	0.76	0.77	0.75	44.83
Item 8	Absorption	0.81	0.83	0.84	57.54
Item 9	Absorption	0.69	0.76	0.73	33.19

CFA Goodness-of-Fit Statistics

None of the Three Models Achieved Good Fit



Goodness-of-Fit: Full Table with Thresholds

Table 5 | Red = Failed to Meet Acceptable Threshold

Fit Statistic	One-Factor	Two-Factor	Three-Factor	Good Fit Threshold
Chi2 (df)	633.90 (27)	354.49 (26)	247.76 (24)	Non-significant
RMSEA	**0.181**	**0.192**	**0.167**	< 0.05 (pref.) / < 0.08
RMSEA 90% CI	0.169-0.194	0.175-0.192	0.154-0.180	--
AIC	16221.47	8246.29	8143.56	Lower = better
BIC	16343.70	8353.66	8258.60	Lower = better
CFI	**0.895**	**0.882**	**0.920**	> 0.95
TLI	**0.860**	**0.837**	**0.880**	> 0.95
SRMR	0.046	0.049	0.065	< 0.08

Red values = Failed to meet acceptable fit thresholds. Only SRMR met the < 0.08 criterion for all models.

Key Findings Summary

EFA vs. CFA Results

EFA Results

One factor explains > 70% variance

KMO = 0.922 (Marvellous)

All loadings: 0.65 - 0.93

MAP test: 1-factor

Forced 3-factor: items misload

CFA Results

RMSEA: 0.167 - 0.192 (poor)

CFI: 0.882 - 0.920 (below 0.95)

TLI: 0.837 - 0.880 (below 0.95)

SRMR: 0.046 - 0.065 (acceptable)

No model shows good overall fit

Interpretation

EFA supports 1-factor structure

CFA cannot confirm any model

3 subscales highly correlated

Possible gender/culture effects

Further research needed

Discussion

Why Did CFA Fail to Confirm the EFA Results?

Possible Explanations:

1. Gender Effects: All-female sample may experience work engagement differently than mixed-gender samples used in original UWES validation

2. Cultural Context: Swedish work culture emphasizes egalitarianism and work-life balance, potentially affecting how engagement manifests

3. Subscale Overlap: Inter-factor correlations of 0.79-0.84 suggest the three dimensions are not clearly distinct -- they may be measuring the same thing

4. Instrument Limitation: Shirom (2003) argued the three UWES dimensions were not theoretically deduced and overlap conceptually

Comparison with Similar Studies:

- Wefald et al. (2012): Similar sample size (n = 382), similar education level, RMSEA = 0.18 and 0.16 -- almost identical to our 0.181 and 0.167
- This suggests the issue may be with the UWES-9 instrument itself, not specific to this sample

Women & Work Engagement

Implications for Workforce Policy & Research

What This Study Tells Us About Women's Engagement:

- Women in this sample reported moderately high engagement (Mean = 4.06/6.0)
- Dedication (meaning, pride, enthusiasm) scored highest at 4.24
- Item 7 'I am proud of the work that I do' had the highest individual score (4.62)
- This suggests pride and purpose are strongest drivers for women's engagement

Practical Implications:

- Organizations should focus on meaningfulness and purpose to engage women
- Standard 3-subscale UWES-9 scoring may not be appropriate for all-female populations
- Total score (1-factor) may be more reliable than subscale scores for women
- Work engagement instruments may need gender-specific validation

Limitations:

- No data on specific occupations (only education level)
- Swedish-speaking women only (though 21.6% had immigrant background)
- Age range limited to 26-37 years
- University-educated majority (61%) may not represent all working women

Technical Improvements to the UWES-9 Framework

Proposed Methodological Enhancements

Bifactor Modeling

Since the three subscales (Vigor, Dedication, Absorption) are highly correlated (0.79-0.84), a Bifactor Model would allow for:

- A general 'Engagement' factor accounting for shared variance across all 9 items
- Specific sub-factors capturing unique variance of each subscale
- Better separation of general vs. specific engagement components
- This addresses the core problem: are the subscales truly distinct?

Multigroup CFA & Invariance Testing

Future research should use Multigroup CFA to determine if the factor structure is invariant across:

- Gender: Compare all-female vs. mixed-gender samples
- Nationality: Test across Swedish, Norwegian, Finnish populations
- Occupation: Compare engagement structure across professions
- This would reveal whether poor fit is due to instrument or sample

Conclusion

Study Summary:

- EFA on n = 341 women: One-factor solution best fits the data
(KMO = 0.922, > 70% variance explained, all loadings 0.65-0.93)

- CFA on n = 342 women: Could NOT confirm any model (1-, 2-, or 3-factor)
(RMSEA never below 0.167, CFI/TLI below 0.95 for all models)

Key Takeaway:

The optimal factor structure of the UWES-9 remains unresolved.
Further research is needed to understand how gender, nationality,
and occupation influence the measurement of work engagement.

Future Directions:

- Test UWES-9 in larger all-female samples across different cultures
- Develop gender-specific engagement instruments
- Investigate whether a shorter version (UWES-3) performs better
- Examine the role of occupation type in factor structure

Factor Analysis Methods Applied

EFA & CFA Techniques Used in This Study

Method	Purpose	Key Finding
KMO Test	Sampling adequacy	0.922 (Marvellous)
Bartlett's Test	Sphericity	$p < 0.001$ (Significant)
EFA (ML extraction)	Identify factor structure	1 factor, > 70% variance
Scree Plot	Visual eigenvalue analysis	Supports 1-factor
Velicer's MAP	Minimum average partial	Supports 1-factor
Parallel Analysis	Compare with random data	Suggested 4 factors
CFA: 1-Factor	Test unidimensional model	RMSEA = 0.181 (poor)
CFA: 2-Factor	Test V+D / A split	RMSEA = 0.192 (poor)
CFA: 3-Factor	Test V / D / A structure	RMSEA = 0.167 (poor)
Cronbach's Alpha	Internal consistency	0.947 (Excellent)

References

Primary Study:

Willmer, M., Westerberg Jacobson, J. & Lindberg, M. (2019). Exploratory and Confirmatory Factor Analysis of the 9-Item UWES in a Multi-Occupational Female Sample. Frontiers in Psychology, 10, 2771. DOI: 10.3389/fpsyg.2019.02771

Supporting References:

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Kulikowski, K. (2017). Do we all agree on how to measure work engagement? Factorial validity of UWES-9 and UWES-17. Polish Psychological Bulletin, 48(3), 350-360.

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Thank You!

Questions?

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